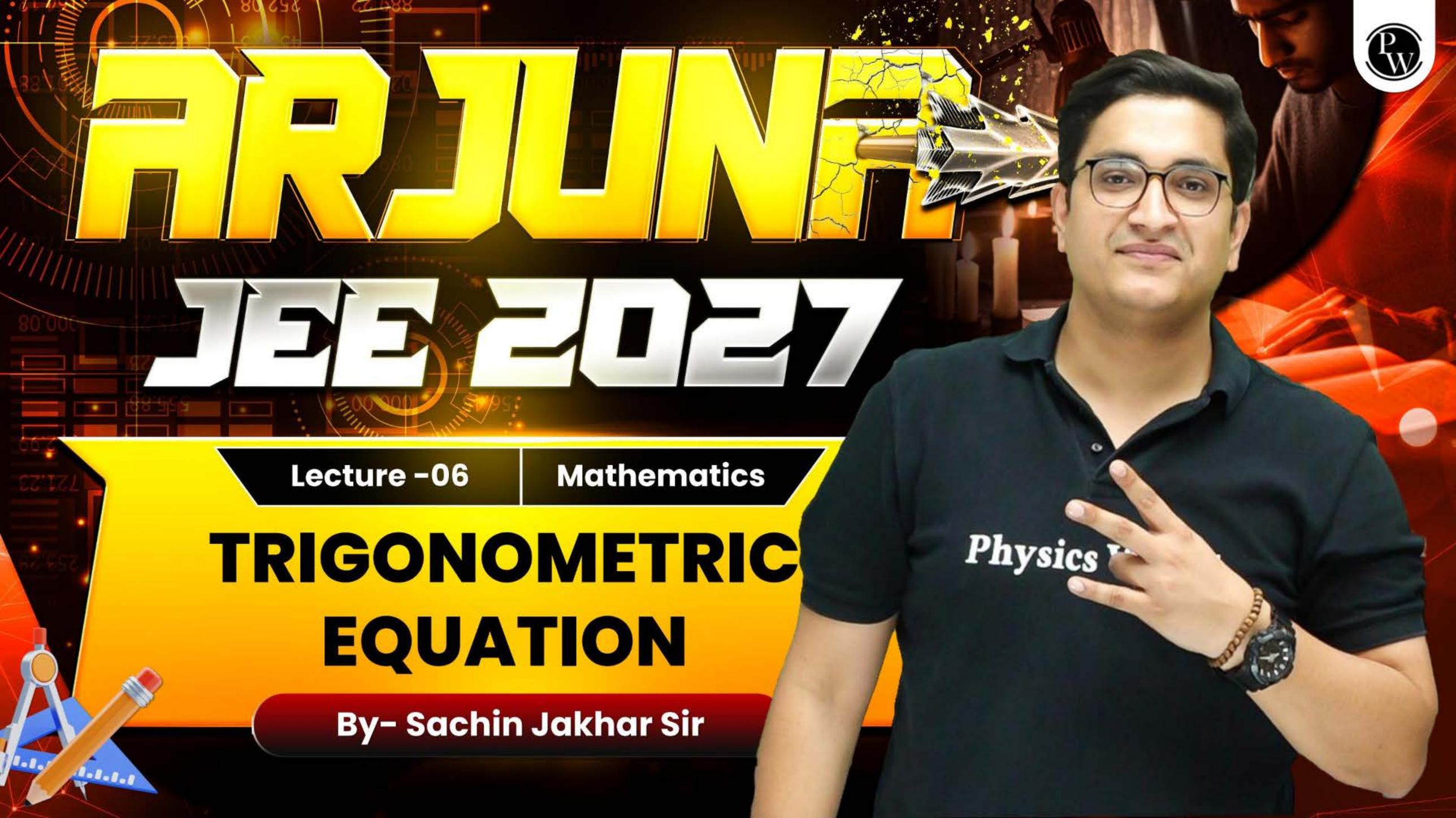
ARJUNA JEE 2027

Lecture -06 | Mathematics

TRIGONOMETRIC EQUATION

By- Sachin Jakhar Sir



Topics *to be covered*

A **Types of Trigonometric Equations (T.E.)**

B **Graphically Solving T.E.**

C **Question Practice**

D



H.W. Discussion

Let $S = \{\theta \in [0, 2\pi) : \tan(\pi\cos\theta) + \tan(\pi\sin\theta) = 0\}$. Then $\sum_{\theta \in S} \sin^2\left(\theta + \frac{\pi}{4}\right)$ is equal to

$\tan(\pi\cos\theta) = \tan(-\pi\sin\theta)$

$\theta = n\pi + \alpha$ ... (I)

$\pi\cos\theta = n\pi - \pi\sin\theta$

$\sin\theta + \cos\theta = n$

$[-\sqrt{2}, \sqrt{2}]$ $-1, 0, 1$.

$n = -1$:

$\sin\theta + \cos\theta = -1$ ✓

$\div\sqrt{2}$

$\frac{1}{\sqrt{2}}\sin\theta + \frac{1}{\sqrt{2}}\cos\theta = -\frac{1}{\sqrt{2}}$

$\cos\frac{\pi}{4}$ $\sin\frac{\pi}{4}$

$\sin\left(\theta + \frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}} = \sin\left(\frac{-\pi}{4}\right)$

Sumit:

$\theta + \frac{\pi}{4} = m\pi + (-1)^m\left(\frac{-\pi}{4}\right)$

Santosh.

no. of soln. = 2 soln.

$\theta + \frac{\pi}{4} = \pi$, $\theta = \frac{3\pi}{4}$

$\theta + \frac{\pi}{4} = \frac{5\pi}{4}$, $\theta = \pi$

$\theta + \frac{\pi}{4} = \frac{7\pi}{4}$, $\theta = \frac{3\pi}{2}$

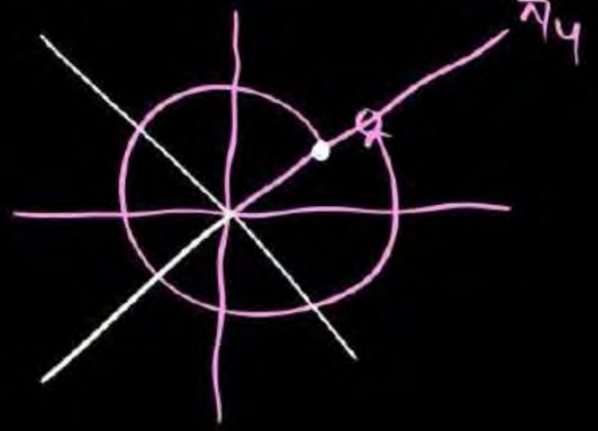
$\frac{1}{2} + \frac{1}{2} + 0 + 0 + \frac{1}{2} + \frac{1}{2}$

2 Ans.

$\theta \in [0, 2\pi)$

$\sin\left(\theta + \frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}$

$\theta + \frac{\pi}{4} \in \left[\frac{\pi}{4}, 2\pi + \frac{\pi}{4}\right)$



$n = 0$

$\sin\theta + \cos\theta = 0$

$\div\sqrt{2}$

$\sin\left(\theta + \frac{\pi}{4}\right) = 0$

↳ no. of soln $= 2$

$n = 1$

$\sin\theta + \cos\theta = 1$

$\div\sqrt{2}$

$\sin\left(\theta + \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$

↳ no. of soln $= 2$

$\theta + \frac{\pi}{4} = \frac{\pi}{4}$ → $\theta = 0$

$\theta + \frac{\pi}{4} = \frac{3\pi}{4}$ → $\theta = \frac{\pi}{2}$

Question

General solution of $\sin x + \sin 5x = \sin 2x + \sin 4x$ is

A $\frac{n\pi}{3}$

B $\frac{2n\pi}{3}$

C $2n\pi$

D $n\pi$

$2\sin 3x \cos 2x - 2\sin 3x \cos x = 0$

$2\sin 3x\,(\cos 2x - \cos x) = 0$

$\sin 3x = 0$ ∪ $\cos 2x = \cos x$

$3x = n\pi$

$x = \frac{n\pi}{3}$

$2x = 2n\pi \pm x$

+ : $2x = 2n\pi + x$ ⇒ $x = 2n\pi$

OR

− : $2x = 2n\pi - x$ ⇒ $3x = 2n\pi$ ⇒ $x = \frac{2n\pi}{3}$

Question

General solution of $\sin x + \sin 5x = \sin 2x + \sin 4x$ is

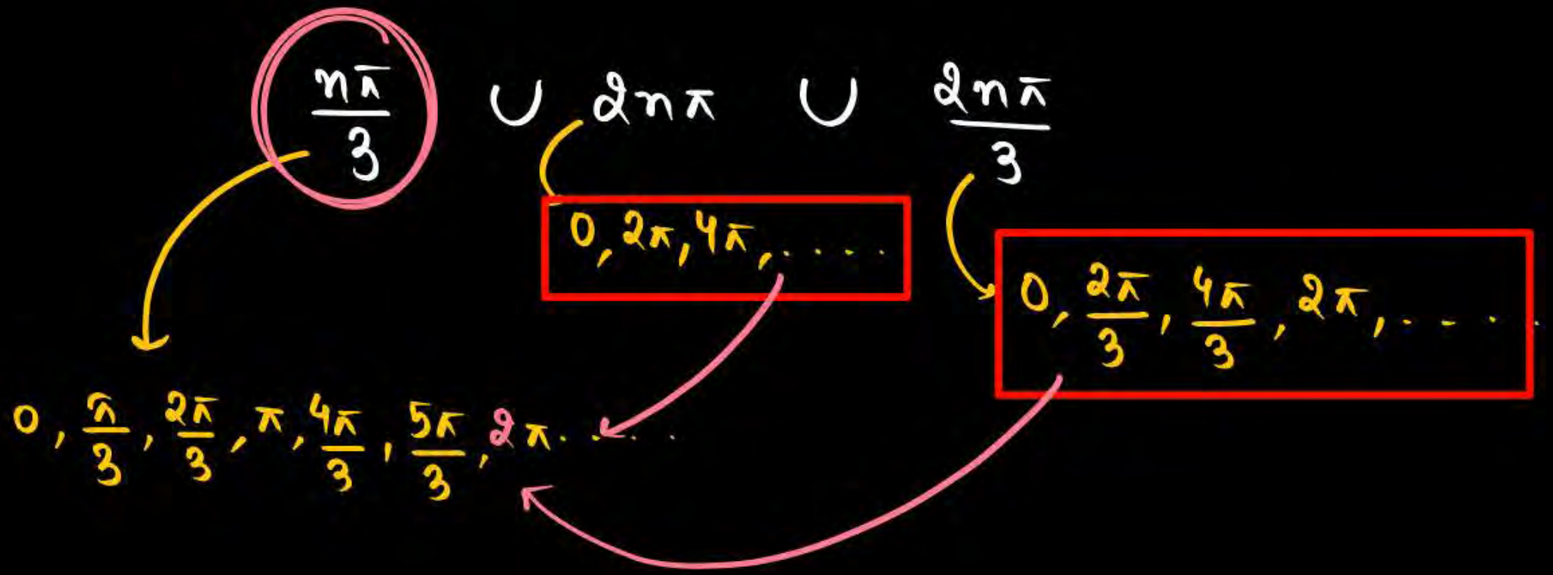
75%

A $\frac{n\pi}{3}$

B $\frac{2n\pi}{3}$

C $2n\pi$

D $n\pi$



QUESTION

Number of solutions of the trigonometric equation in $[0, \pi]$, is $\sin 3\theta = 4\sin\theta \cdot \sin 2\theta \cdot \sin 4\theta$.

(A) 4

(B) 16

(C) 8

(D) 0

72%

$\sin\theta(3-4\sin^2\theta) - 4\sin\theta \cdot \sin 2\theta \sin 4\theta = 0$

$\sin\theta = 0$

$\theta = 0, \pi$

$\sin\theta\left\{3-4\sin^2\theta - 2(2\sin 2\theta \sin 4\theta)\right\} = 0$ (B, A)

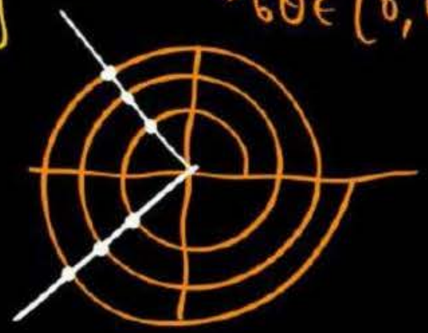
$\cos 6\theta = -\frac{1}{2} \longrightarrow$ 6 soln.

$\sin\theta\left\{3-2+2\cos 2\theta - 2\cos(2\theta) + 2\cos(6\theta)\right\} = 0$

$\theta \in [0, \pi]$

$6\theta \in [0, 6\pi]$

$(1+2\cos 6\theta)\sin\theta = 0$



Question

Find value of λ for which equation $\sin^6 x + \cos^6 x = \lambda - 1$ is consistent.

$$\sin^6 x + \cos^6 x = 1 - 3(\sin x \cos x)^2$$

$$= 1 - 3\left(\frac{\sin 2x}{2}\right)^2$$

$$= 1 - \frac{3}{4}\sin^2 2x$$

\# $\sin 2x \in [-1, 1]$

$\sin^2 2x \in [0, 1]$

$-\frac{3}{4}\sin^2 2x \in [-\frac{3}{4}, 0]$

$1 - \frac{3}{4}\sin^2 2x \in [\frac{1}{4}, 1]$

$\lambda - 1 \in [\frac{1}{4}, 1]$

$\lambda \in [\frac{5}{4}, 2]$

fnf

Question [JEE Mains-2020]

If the equation $\cos^4\theta + \sin^4\theta + \lambda = 0$ has real solutions for θ, then λ lies in the interval

$1 - 2\sin^2\theta\cos^2\theta = -\lambda$

A $[-3/2, -5/2]$

B $[-1/2, -1/4]$

C $[-5/4, -1]$

D $[-1, -1/2]$

QUESTION [JEE Mains-2020]

79%

If the equation $\cos^4\theta + \sin^4\theta + \lambda = 0$ has **real solutions** for θ, then λ lies in the interval

(A) $[-3/2, -5/2]$

(B) $[-1/2, -1/4]$

(C) $[-5/4, -1]$

(D) $[-1, -1/2]$

$$\cos^4\theta + \sin^4\theta = -\lambda$$

RHS $= -\lambda \in$ Range of LHS

*** $f(\theta) = \cos^4\theta + \sin^4\theta$

$= 1 - 2(\sin\theta\cos\theta)^2$

$= 1 - 2\left(\frac{\sin 2\theta}{2}\right)^2 = 1 - \frac{1}{2}\sin^2 2\theta$

$\sin 2\theta \in [-1, 1]$

$\sin^2 2\theta \in [0, 1]$

$-\frac{1}{2}\sin^2 2\theta \in \left[-\frac{1}{2}, 0\right]$

$1 - \frac{1}{2}\sin^2 2\theta \in \left[\frac{1}{2}, 1\right]$

$-\lambda \in \left[\frac{1}{2}, 1\right]$

$\lambda \in \left[-1, -\frac{1}{2}\right]$

Question [JEE (Adv.)-2014]

For $x \in (0, \pi)$, the equation $\sin x + 2\sin 2x - \sin 3x = 3$ has



A Infinitely many solutions

B Three solutions

C One solution

D No solution

Challenger Question (New to Me)

$f \nrightarrow f: x \in \left\{(2n+1)\frac{\pi}{2}\right\}, n \in I$.

Solve : ~~(i)~~ $\sin^4 x = 1 + \cos^8 x$

(ii) $\sin^4 x + \cos^{12} x = 1$

(iii) $\sin^{14} x + \cos^{31} x = 1$

(i) Range Comp.

LHS $\begin{cases} \sin^2 x \in [0,1] \\ \sin^4 x \in [0,1] \end{cases}$

RHS $\begin{cases} \cos^2 x \in [0,1] \\ \cos^4 x \in [0,1] \\ 1+\cos^8 x \in [1,2] \end{cases}$

equality will hold when LHS = RHS = 1

$\sin^4 x = 1$

$\Downarrow$

$\sin x = \pm 1$

$1 + \cos^8 x = 1$

$\cos^8 x = 0$

$\cos x = 0$ ✓

(i) Santosh:

$(\sin^2 x)^2 = 1 + \cos^8 x$

$(1 - \cos^2 x)^2 = 1 + \cos^8 x$

$\vdots$

Challenger Question (New to Me)

Solve : (i) $\sin^4 x = 1 + \cos^8 x$

(ii) $\sin^4 x + \cos^{12} x = 1$

(iii) $\sin^{14} x + \cos^{31} x = 1$

feel.

$\sin x = \pm 1 \cap \cos x = 0$ — (I)

$\cup$

$\sin x = 0 \cap \cos x = \pm 1$ — (II)

$\sin^4 x \le \sin^2 x$

$\cos^{12} x \le \cos^2 x$

add.

$\sin^4 x + \cos^{12} x \le 1.$

equality hold :

$\sin^4 x = \sin^2 x \cap \cos^{12} x = \cos^2 x$

$\sin x = 0, 1, -1.$

$\sin^4 x + \cos^{12} x = 1$.

$\sin^4 x = 0 \Rightarrow \cos^{12} x = 1$.

$\sin x = 0 \cap \cos x = \pm 1$ $\Big\}$ $x = n\pi$.

$\cup$

$\sin^4 x = 1 \Rightarrow \cos^{12} x = 0$.

$\sin x = \pm 1 \cap \cos x = 0$ $\Big\}$ $x = (2n+1)\frac{\pi}{2}$

(ii)

$\sin^4 x \le \sin^2 x$

$\cos^{12} x \le \cos^2 x$

add. ——————

$\sin^4 x + \cos^{12} x \le 1$.

equality hold:

$\sin^4 x = \sin^2 x \cap \cos^{12} x = \cos^2 x$

$\sin x = 0, 1, -1$.

$\sin^4 x = 0, 1$.

$\therefore x \in \{n\pi\} \cup \left\{(2n+1)\frac{\pi}{2}\right\}$

$\alpha^{\text{odd}_1} = \alpha^{\text{odd}_2}$

$\alpha = 0, 1, -1$

ex:- $\alpha^{17} = \alpha^{11}$

$\alpha^{17} - \alpha^{11} = 0$

$\alpha^{11}(\alpha^6 - 1) = 0$

$\alpha = 0, \quad \alpha = \pm 1$

$\alpha^{\text{even}_1} = \alpha^{\text{even}_2}$

$\alpha = 0, 1, -1$ ✓

ex:- $\alpha^{54} = \alpha^{12}$

$\alpha^{54} - \alpha^{12} = 0$

$\alpha^{12}(\alpha^{42} - 1) = 0$

$\alpha = 0$ $\quad$ $\alpha^{42} = 1$

$\alpha = \pm 1$

$\alpha^{\text{odd}} = \alpha^{\text{even}}$

$\alpha = 0, 1.$

ex:- $\alpha^7 = \alpha^4$

$\alpha^7 - \alpha^4 = 0$

$\alpha^4(\alpha^3 - 1) = 0$

$\alpha = 0$ $\quad$ $\alpha^3 = 1.$

$\alpha = 1.$

Challenger Question (New to Me)

Solve : (i) $\sin^4 x = 1 + \cos^8 x$

(ii) $\sin^4 x + \cos^{12} x = 1$

(iii) $\sin^{14} x + \cos^{31} x = 1$

feel :

$\sin^{14} x = 1 \cap \cos^{31} x = 0$

$\sin x = \pm 1$, $\cos x = 0$

$\sin^{14} x = 0 \cap \cos^{31} x = 1$

$\sin x = 0$, $\cos x = 1$

$x = (2n+1)\frac{\pi}{2} \cup x = (2n\pi)$

fnf

Acidic Question

Solve: $\sin^{23}x + \cos^{77}x = 1$

$\sin^{23}x = 1 \cap \cos^{77}x = 0$

$\sin x = 1$, $\cos x = 0$

$\Rightarrow x = (4n+1)\frac{\pi}{2}$

$\sin^{23}x = 0 \cap \cos^{77}x = 1$

$\sin x = 0$, $\cos x = 1$

$\Rightarrow x = 2n\pi$

$\cup$

Question [JEE Mains-2021]

The number of solution of $\sin^7 x + \cos^7 x = 1$, $x \in [0, 4\pi]$ is equal to

A 11

B 7

C 5

D 9



Critical Thinking Question [JEE Mains-2019]

If $\sin^4\alpha + 4\cos^4\beta + 2 = 4\sqrt{2}\sin\alpha\cos\beta$, $\alpha, \beta \in [0, \pi]$, then $\cos(\alpha+\beta) - \cos(\alpha-\beta)$ is equal to

A $\sqrt{2}$

B $-\sqrt{2}$

C -1

D 0

AM ≥ GM.

$$\frac{\sin^4\alpha + 4\cos^4\beta + 1 + 1}{4} \geq \left(\sin^4\alpha \cdot 4\cos^4\beta \cdot 1 \cdot 1\right)^{\frac{1}{4}}$$

$$4^{\frac{1}{4}} = \sqrt{2}$$

$$\left(2^2\right)^{\frac{1}{4}}$$

$$\sin^4\alpha + 4\cos^4\beta + 2 \geq 4\sqrt{2}\sin\alpha\cdot\cos\beta$$

equality hold:

$$\sin^4\alpha = 4\cos^4\beta = 1 = 1$$

$$\sin^4\alpha = 1 \quad \& \quad \cos^4\beta = \frac{1}{4}$$

$$\sin\alpha = \pm 1$$

$$\alpha = \frac{\pi}{2}$$

$$\cos\beta = \pm\frac{1}{\sqrt{2}}$$

$$\beta = \frac{\pi}{4} \text{ or } \frac{3\pi}{4}$$

$$\cos\left(\frac{\pi}{2}+\frac{\pi}{4}\right) - \cos\left(\frac{\pi}{2}-\frac{\pi}{4}\right)$$

$$-\sin\frac{\pi}{4} - \sin\frac{\pi}{4}$$

$$-2\times\frac{1}{\sqrt{2}}$$

Critical Thinking Question [JEE Mains-2019]

If $\sin^4\alpha + 4\cos^4\beta + 2 = 4\sqrt{2}\sin\alpha\cos\beta$, $\alpha, \beta \in [0, \pi]$, then $\cos(\alpha+\beta) - \cos(\alpha-\beta)$ is equal to

A $\sqrt{2}$

B $-\sqrt{2}$

C -1

D 0

$\sin\alpha > 0$

$\cos\beta \rightarrow \oplus$ ✓ / $\ominus$ × / 0 ×

if $\sin\alpha = 0$

$\Rightarrow 4\cos^4\beta + 2 = 0$ } n.p.

if $\cos\beta = 0$

$\sin^4\alpha + 2 = 0$ } n.p.

TYPE-06(B)

When different types of variables or functions are present.

Blue बादल (KKPNCC)

Same type or variety of functions को ek side रखों, now compare range of LHS & RHS.

Question [JEE Mains-2022]

The number of elements in the set $S = \left\{x \in \mathbb{R} : 2\cos\left(\frac{x^2+x}{6}\right) = 4^x + 4^{-x}\right\}$

T.F.

$2\cos\left(\frac{x^2+x}{6}\right) \leq 2$

$4^x + 4^{-x} \geq 2$

E.F.

A 1

B 3

C 0

D Infinite

* equality hold

LHS = RHS = 2

LHS = 2

$2\cos\left(\frac{x^2+x}{6}\right) = 2$

$\cos\left(\frac{x^2+x}{6}\right) = 1$

$\frac{x^2+x}{6} = 0$

$x = 0, -1.$

$4^x = \frac{1}{4^x}$

$x = 0$

$x = 0$

LHS $= 2\cos\left(\frac{x^2+x}{6}\right)$

$\cos\left(\frac{x^2+x}{6}\right) \in [-1, 1]$

$2\cos\left(\frac{x^2+x}{6}\right) \in 2 \times [-1, 1]$

$\in [-2, 2]$

RHS $= 4^x + \frac{1}{4^x}$

AM $\geq$ GM

$\frac{4^x + \frac{1}{4^x}}{2} \geq (1)^{\frac{1}{2}}$

$4^x + \frac{1}{4^x} \geq 2$

Question [JEE Mains-2024]

The sum of the solutions $x \in R$ of the equation

$$\frac{3\cos 2x+\cos^3 2x}{\cos^6 x-\sin^6 x} = x^3 - x^2 + 6$$ is

T.f.

cubic.

A 0

B 1

C −1

D 3

$$LHS = \frac{\cos^2 x\,(3+\cos^2 2x)}{(\cos 2x)(1-\sin^2 x\cos^2 x)}$$

$$= 4$$

$3+1-\sin^2 2x$.

$4 - 4\cos^2 x \sin^2 x$

$4(1-\cos^2 x \cdot \sin^2 x)$

$\cos^6 x - \sin^6 x$

$= (\cos^2 x)^3 - (\sin^2 x)^3$

$= (\cos^2 x - \sin^2 x)(\cos^4 x + \sin^4 x + \cos^2 x \sin^2 x)$

$1 - 2\sin^2 x \cos^2 x + \cos^2 x \cdot \sin^2 x$

$1 - \sin^2 x \cdot \cos^2 x$.

Question [JEE Mains-2024]

The sum of the solutions $x \in R$ of the equation

$$\frac{3\cos 2x+\cos^3 2x}{\cos^6 x-\sin^6 x} = x^3 - x^2 + 6$$ is

A 0

B 1

C –1

D 3

$4 = x^3 - x^2 + 6$

$x^3 - x^2 + 2 = 0$

RRT. $2 \to \pm 1, \pm 2$

$x = -1$

$(x+1)(x^2 - 2x + 2) = 0$

$D < 0 \to \phi$

$(x+1) = 0$

$x = -1$.

Question

Solve for x and y :

(i) $12 \sin x - 2y^2 = 21 - 8y - 5 \cos x$

(ii) $x^2 + 2x \sin(xy) + 1 = 0$

new.

$12\sin x + 5\cos x = 2y^2 - 8y + 21$.

$x^2 + 1 = -2x \sin(xy)$

$x + \frac{1}{x} = -2\sin(xy)$.



SOLVING GRAPHICALLY



Point-01 : For graphically solving or finding number of solution of Equation : $f(x) = g(x)$

Step-01 : Simplify both side (whose graphs are known or easy to draw)

Step-02 : Draw $y = f(x)$ & $y = g(x)$ on same axis.

Step-03 : Number of solutions = Number of Point of intersection of graph of $f(x)$ & $g(x)$.

Solutions of equation are 'x-coordinate of point of intersection'

Bumper Question

Find number of solutions for following equations:

(i) $\sin x = x$

(ii) $\cos x = x$

(iii) If $x \in [-\pi, \pi]$ & $\sin x = \frac{\sqrt{13}}{15}$

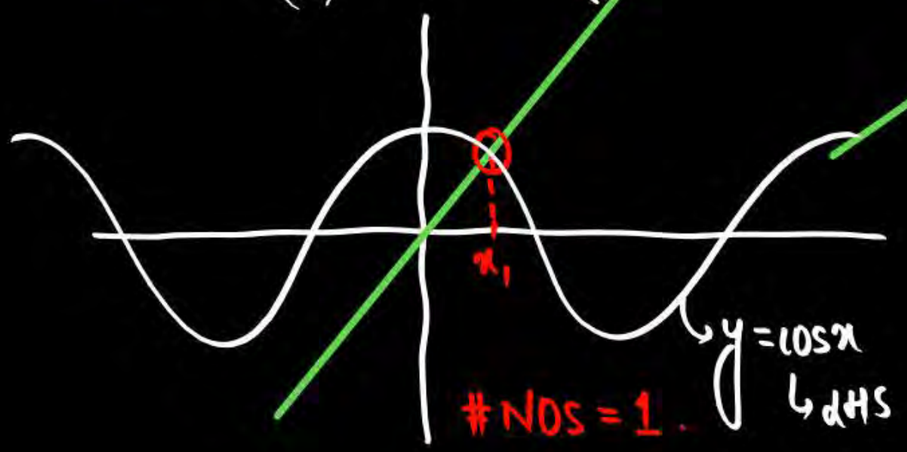
(iv) $\sin x = |x - 2|$

(v) $\sin x = \frac{x}{10}$

(i) $\underbrace{y = \sin x}_{\text{LHS}}$ & $\underbrace{y = x}_{\text{RHS}}$



(ii)



Bumper Question

Find number of solutions for following equations:

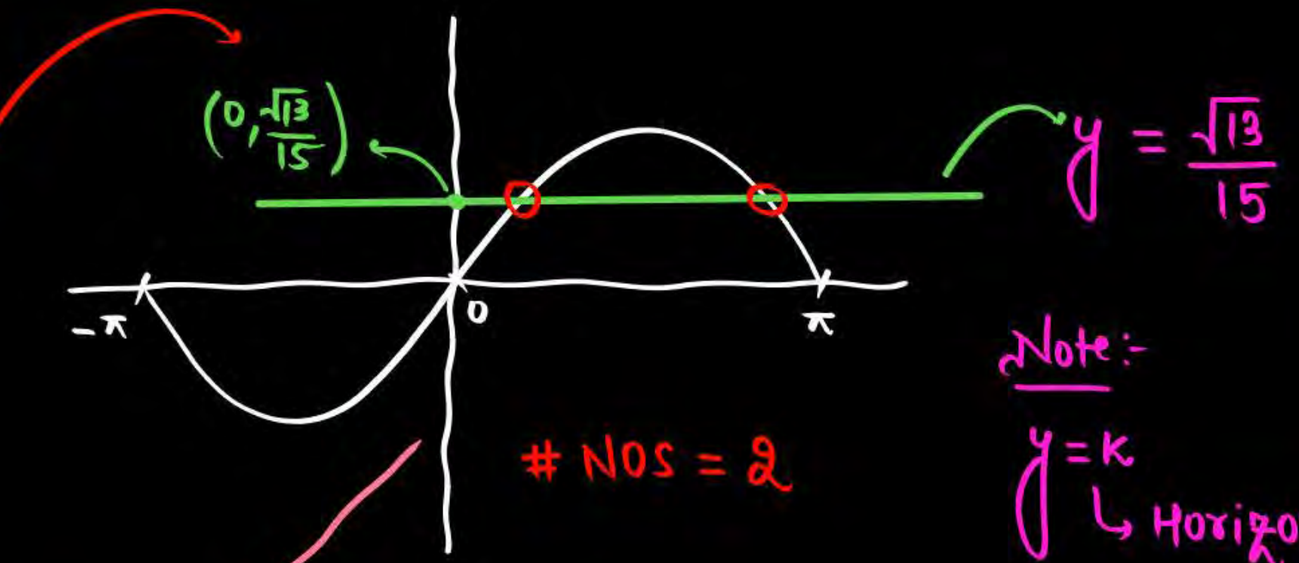
(i) $\sin x = x$

(ii) $\cos x = x$

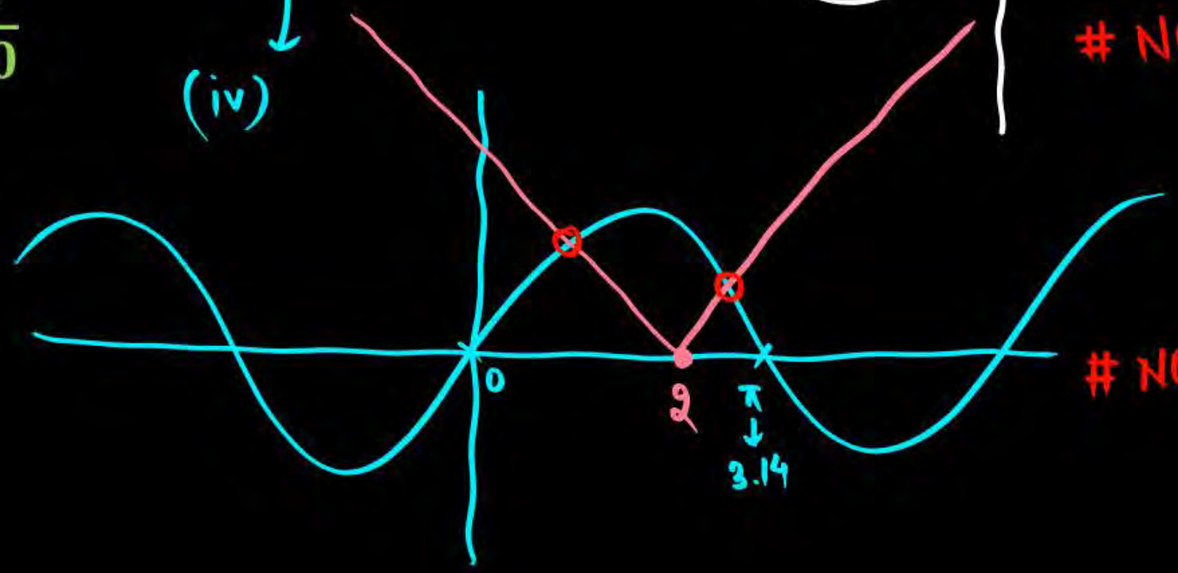
(iii) If $x \in [-\pi, \pi]$ & $\sin x = \frac{\sqrt{13}}{15}$

(iv) $\sin x = |x - 2|$

(v) $\sin x = \frac{x}{10}$



NOS = 2



NOS = 2

Note:-

$y = K$

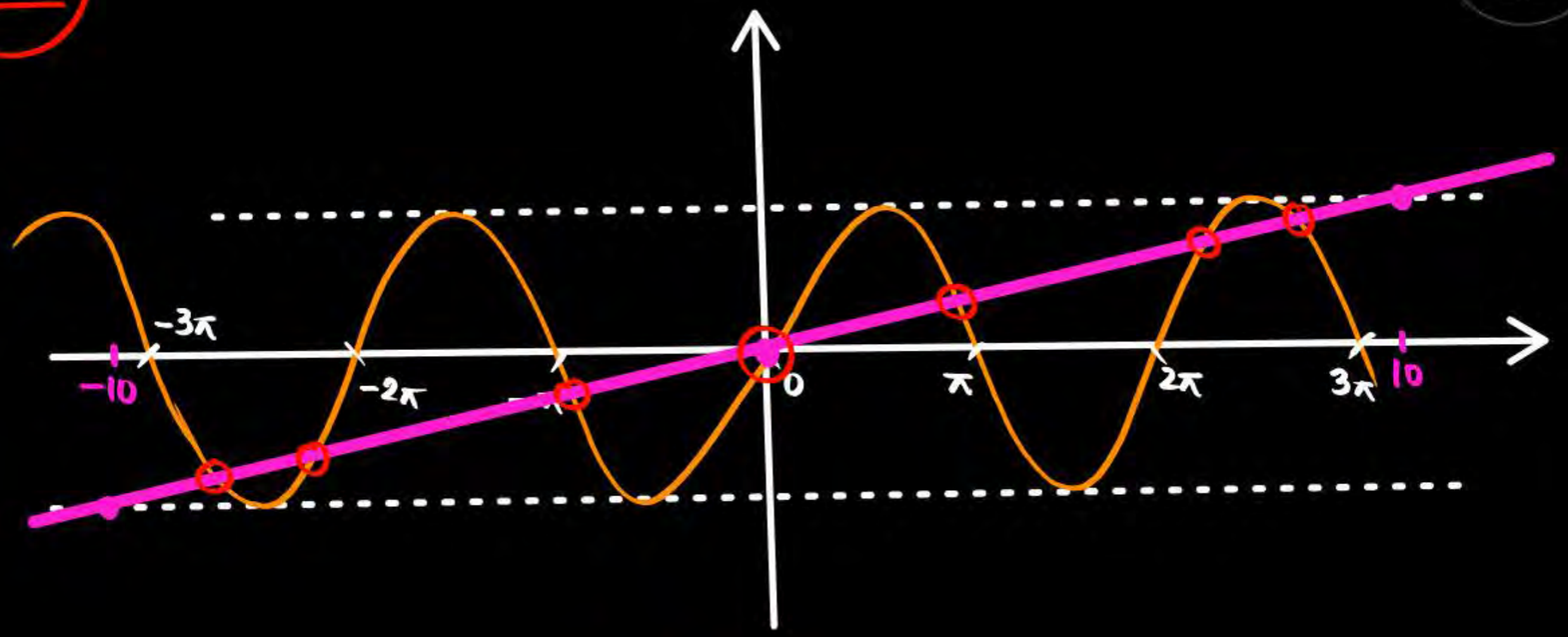
↳ Horizontal lines ($\parallel^{el}$ to x-axis) passing thru $(0, K)$ on y-axis.

(v) $10\sin x = x$

Note

$\sin x = \frac{x}{10}$

$y = \sin x$. $y = \frac{x}{10}$



NOS = 7 .

Question Practice

DIBY-A

If $0 < \theta, \phi < \frac{\pi}{2}$, $x = \sum_{n=0}^{\infty} \cos^{2n}\theta$, $y = \sum_{n=0}^{\infty} \sin^{2n}\phi$ and $z = \sum_{n=0}^{\infty} \cos^{2n}\theta \cdot \sin^{2n}\phi$ then.

[JEE Mains-2021]

(A) $xyz = 4$

(B) $xy - z = (x + y)z$

(C) $xy + yz + zx = z$

(D) $xy + z = (x + y)z$

[Ans. : (D)]

DIBY-B

Let S be the sum of all solutions (in radians) of the equation $\sin^4\theta + \cos^4\theta - \sin\theta\cos\theta = 0$ in $[0, 4\pi]$. Then $\frac{8S}{\pi} =$

[JEE Mains-2021]

[Ans. : 56]

DIBY-C

The number of integral values of 'k' for which the equation $3\sin x + 4\cos x = k + 1$ has a solution, $k \in R$ is

[JEE Mains-2021]

[Ans. : 11]

DIBY-D

The number of solutions of the equation $1 + \sin^4 x = \cos^2 3x$, $x \in \left[-\frac{5\pi}{2}, \frac{5\pi}{2}\right]$ is :

[JEE Mains-2019]

(A) 5 (B) 4 (C) 7 (D) 3

[Ans. : (A)]

DIBY-E

The sum of solutions of the equation $\frac{\cos x}{1+\sin x} = |\tan 2x|$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) - \left\{\frac{\pi}{4}, -\frac{\pi}{4}\right\}$ is:

[JEE Mains-2021]

(A) $-\frac{\pi}{15}$ (B) $\frac{\pi}{10}$ (C) $-\frac{7\pi}{30}$ (D) $-\frac{11\pi}{30}$

[Ans. : (D)]

DIBY-F

The number of values of x in the interval $\left(\frac{\pi}{4}, \frac{7\pi}{4}\right)$ for which $14\text{cosec}^2 x - 2\sin^2 x = 21 - 4\cos^2 x$ holds, is _____.

[JEE Mains-2022]

[Ans. : 4]

DIBY-G

Let $S = \left[-\pi, \frac{\pi}{2}\right) - \left\{-\frac{\pi}{2}, -\frac{\pi}{4}, -\frac{3\pi}{4}, \frac{\pi}{4}\right\}$. Then the number of elements in the set $A = \{\theta \in S : \tan\theta(1 + \sqrt{5}\tan(2\theta)) = \sqrt{5} - \tan(2\theta)\}$ is

[JEE Mains-2022]

[Ans. : 5]

DO IT BY YOURSELF (DIBY)

DIBY-16

$\frac{\sin 3\theta}{2\cos 2\theta+1} = \frac{1}{2}$ if

(A) $\theta = n\pi + \frac{\pi}{6}, n \in I$

(B) $\theta = 2n\pi - \frac{\pi}{6}, n \in I$

(C) $\theta = n\pi + (-1)^n \frac{\pi}{6}, n \in I$

(D) $\theta = n\pi - \frac{\pi}{6}, n \in I$

[Ans. : (C)]

DIBY-17

If $0 \leq x < \frac{\pi}{2}$, then the number of values of x for which $\sin x - \sin 2x + \sin 3x = 0$, is

(A) 2 (B) 1 (C) 3 (D) 4

[Ans. : (A)]

DIBY-18

If the sum of solutions of the system of equations $2\sin^2\theta - \cos 2\theta = 0$ and $2\cos^2\theta + 3\sin\theta = 0$ in the interval $[0, 2\pi]$ is $k\pi$, then k is equal to .

[Ans. : 03]

DIBY-19

Solve : $\sin(\cos x) = \cos(\sin x)$

[Ans. : No Solution]

DIBY-20

Solve : $\operatorname{cosec} x - \operatorname{cosec} 2x = \operatorname{cosec} 4x$

[Ans. : $x = (2n+1)\frac{\pi}{7}$ & $n \neq 7m+3, n \in I$ & $m \in I$]

DIBY-21

The general value of θ satisfying the equation $2\sin^2\theta - 3\sin\theta - 2 = 0$, is

[JEE-1995, 2M]

(A) $n\pi + (-1)^n \frac{\pi}{6}$

(B) $n\pi + (-1)^n \frac{\pi}{2}$

(C) $n\pi + (-1)^n \frac{5\pi}{6}$

(D) $n\pi + (-1)^n \frac{7\pi}{6}$

[Ans. : (D)]

DIBY-22

The general solution of $\sin x - 3\sin 2x + \sin 3x = \cos x - 3\cos 2x + \cos 3x$ is

[JEE-1989, 2M]

(A) $n\pi + \frac{\pi}{8}$ (B) $\frac{n\pi}{2} + \frac{\pi}{8}$ (C) $(-1)^n \frac{n\pi}{2} + \frac{\pi}{8}$ (D) $2n\pi + \cos^{-1}\frac{3}{2}$

[Ans. : (C)]

DIBY-23

The possible values of $\theta \in (0, \pi)$ such that $\sin(\theta) + \sin(4\theta) + \sin(7\theta) = 0$ are

[AIEEE-2011]

(A) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$

(B) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{4\pi}{9}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{8\pi}{9}$

(C) $\frac{\pi}{4}, \frac{5\pi}{12}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$

(D) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{35\pi}{36}$

[Ans. : (B)]



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians